



Solving Various Optimization Problems with **METAHEURISTIC METHODS**

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Optimization Problems Are Everywhere



Logistics / Transportation

Minimize transportation cost or delivery time



Manufacturing

Maximize profit or minimize production time



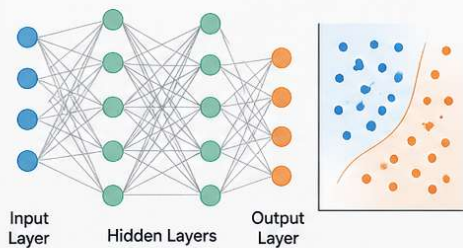
Finance

Maximize return and minimize risk



Machine Learning

Minimize prediction error (optimize model parameters)



Healthcare

Optimize staff scheduling or resource allocation

	MON	TUE	WED	THU	FRI
NURSE A					
NURSE B					
NURSE C					
NURSE D					
DOCTOR A					
DOCTOR B					



Telecommunications

Minimize latency or maximize network capacity



Optimization Model

Optimization problem can be modelled mathematically as

minimize $f(\mathbf{x})$

subject to $g_i(\mathbf{x}) \leq 0$, $i = 1, 2, \dots, m$

$h_j(\mathbf{x}) = 0$, $j = 1, 2, \dots, p$

$l_k \leq x_k \leq u_k$, $k = 1, 2, \dots, n$

- $f(\mathbf{x})$ is the **objective function** to be optimized. Optimization problems are typically written as minimization problems since maximizing $f(\mathbf{x})$ is equivalent to minimizing $-f(\mathbf{x})$.
- $\mathbf{x} = (x_1, x_2, \dots, x_n)$ is a **decision variable** vector. It is feasible if it satisfies all constraints.
- **Constraints** are optional (i.e., the problem may be constrained or unconstrained).
- Solving an optimization problem means finding a feasible decision variable vector $\mathbf{x}$ that minimizes $f(\mathbf{x})$.

Challenges of Classical Optimization Methods

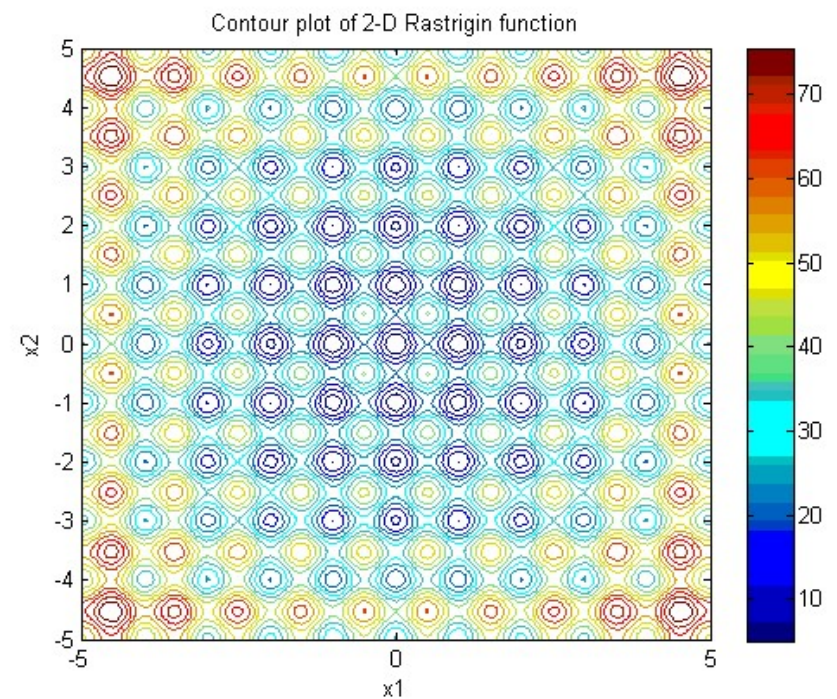
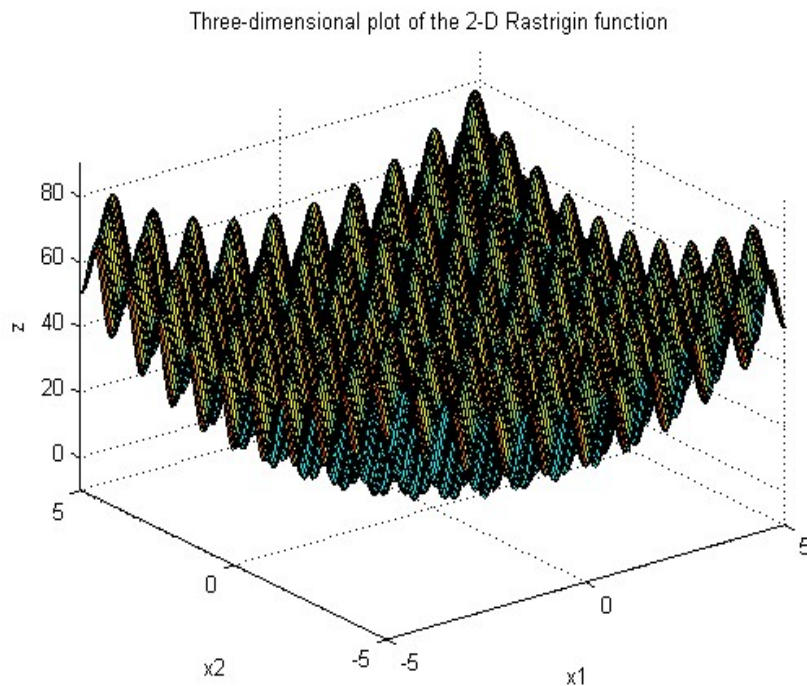
Classical Optimization Methods

- Analytical methods : Find stationary points by solving $\nabla f(x) = \mathbf{0}$.
Use Lagrange multipliers for constrained optimization.
- Iterative methods e.g. Newton's Method, Steepest Descent, Conjugate Gradient, etc

Challenges : These methods often require :

- The objective function to be continuous and differentiable.
- Computing first- or second-order derivatives.
- A good initial solution, since convergence often depends on the starting point.
- For high-dimensional, nonlinear, or multimodal optimization problems, gradient-based methods may converge to a local optimum rather than the global optimum.

Challenges of Classical Optimization Methods



Gradient-based methods may converge to a local minimum, depending on the initial solution.



Metaheuristic Methods



What are Metaheuristic Methods?

Metaheuristic methods are global optimization algorithms that intelligently explore the search space to increase the likelihood of finding the global optimum for complex optimization problems within a reasonable computation time.

Unlike classical optimization methods, metaheuristics:

- **Do not require derivative information**, so it can be applied to nonlinear, nonconvex, nondifferentiable, discontinuous, multimodal, and black-box optimization problems.
- **Can handle** continuous, discrete, and mixed-variable optimization problems.
- Use stochastic search strategies to explore the search space, reducing the risk of becoming trapped in local optima; however, they typically produce **near-optimal solutions** and **do not guarantee the global optimum**.
- **Require algorithm-specific parameter tuning**, which can significantly affect search performance and solution quality.

Diversification and Intensification in Metaheuristics

Metaheuristic algorithms achieve good performance by balancing diversification and intensification throughout the search process.

Diversification (Exploration)	Intensification (Exploitation)
Explores new regions of the search space	Searches thoroughly around promising solutions
Increases population diversity	Improves solution quality
Helps escape local optima	Accelerates convergence
Emphasizes global search	Emphasizes local search

Too much diversification slows convergence. Too much intensification increases the risk of getting trapped in local optima. Successful metaheuristics maintain a balance between the two.

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Spiral Optimization (SpO)

Spiral Optimization (SpO)^[1] is a metaheuristic optimization algorithm that searches for optimal solutions by moving candidate solutions along spiral trajectories toward a promising point in the search space.

The update equation of the SpO algorithm is typically written as

$$\mathbf{x}_i(k+1) = S_n(r, \theta) \mathbf{x}_i(k) - (S_n(r, \theta) - I) \mathbf{x}^*(k)$$

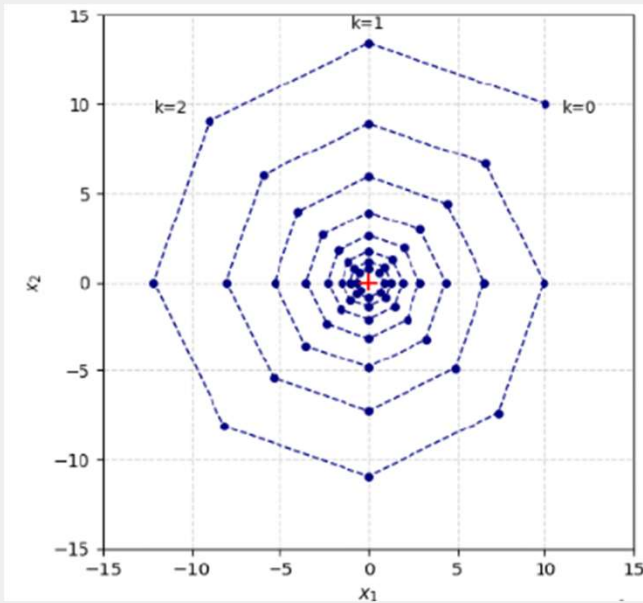
where r controls the convergence speed (smaller r gives faster convergence).

θ controls the rotation angle.

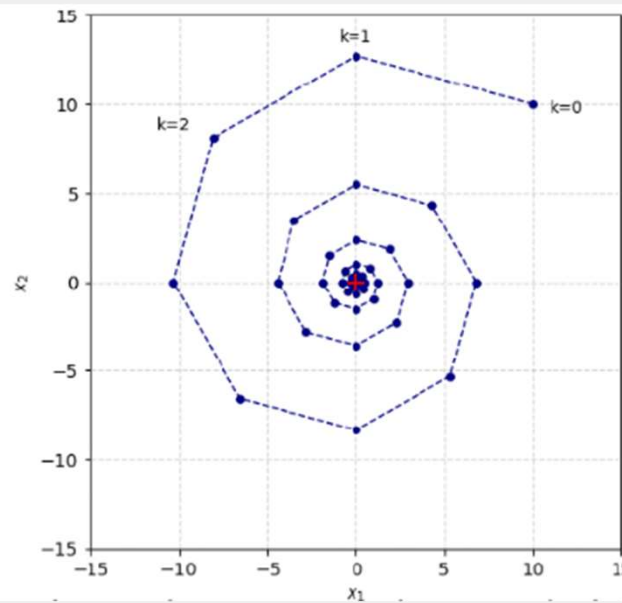
SpO does not introduce random numbers during the iterative search process



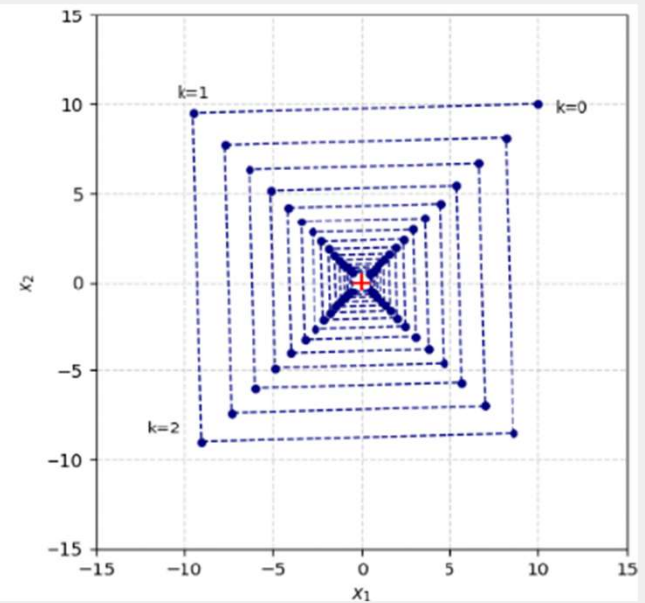
Effect of Spiral Parameters



$$r = 0,95, \quad \theta = \frac{\pi}{4}$$



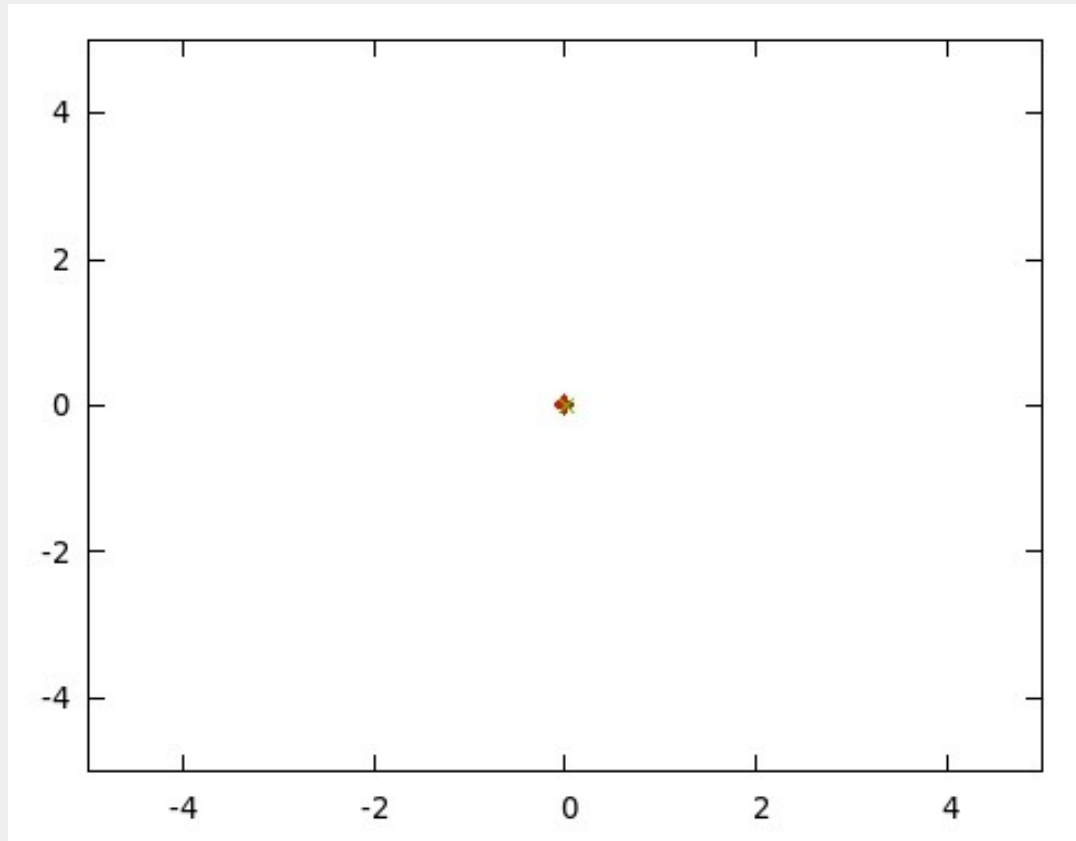
$$r = 0,90, \quad \theta = \frac{\pi}{4}$$



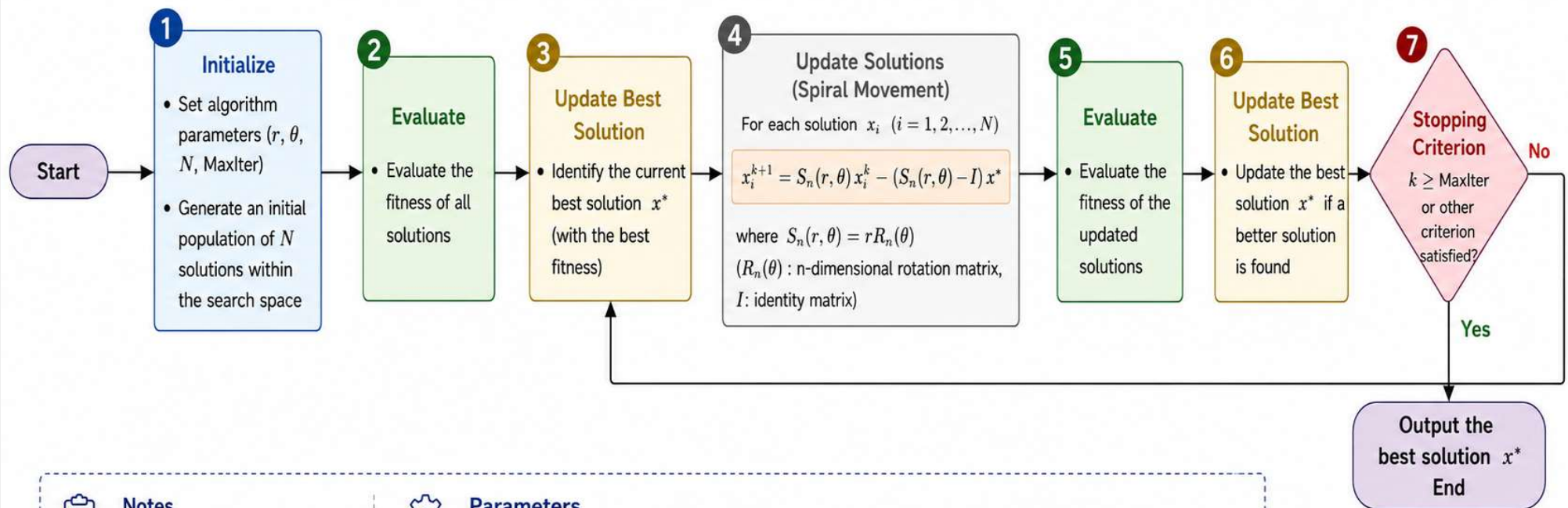
$$r = 0,95, \quad \theta = \frac{\pi}{2}$$

The convergence rate r controls how quickly solutions converge, while the rotation angle θ determines the shape of the spiral trajectory.

Spiral Optimization (SpO) Algorithm



Spiral Optimization (SpO) Algorithm



Notes

- The search is deterministic; no random numbers are used in the iterative process.



Parameters

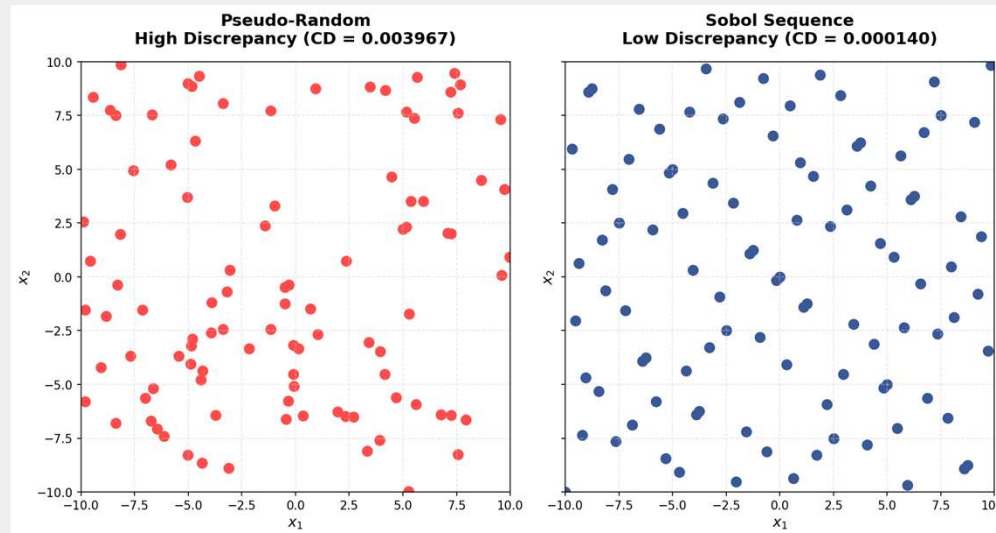
- | | | | |
|--|---|--|--|
| <ul style="list-style-type: none"> r : convergence rate ($0 < r < 1$) | <ul style="list-style-type: none"> θ : rotation angle ($0 < \theta \leq 2\pi$) | <ul style="list-style-type: none"> N : population size | <ul style="list-style-type: none"> MaxIter : maximum number of iterations |
|--|---|--|--|

Initial Population Generation

Since SpO does not generate new random solutions during the search, the quality of the initial population has a significant impact on the search performance.

Pseudo-Random Sampling

- Uneven distribution of initial solutions.
- Some regions may be oversampled, while others remain unexplored.
- May reduce the effectiveness of deterministic search.



Sobol Sequence (Low-Discrepancy)

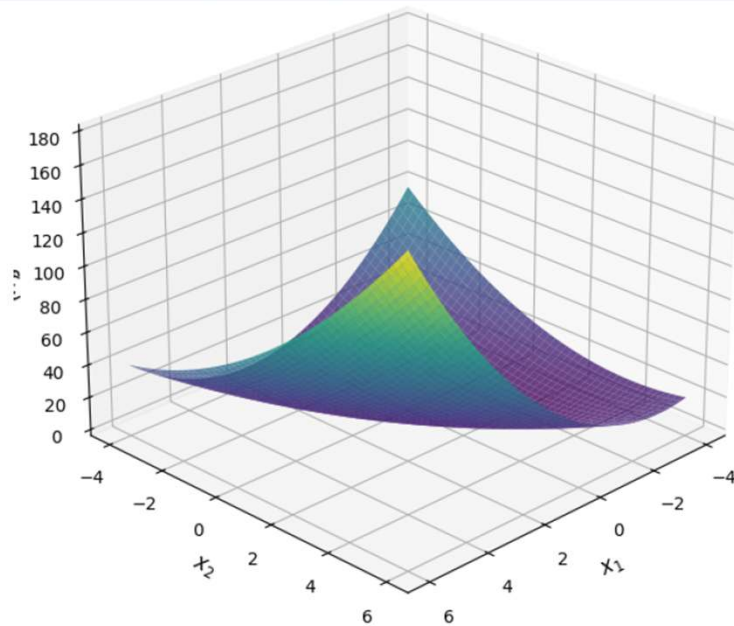
- Uniformly covers the search space.
- Ensures a more representative initial population.
- Improves the effectiveness of deterministic search.

Benchmark test

1. Schwefel (Unimodal)

$$f(\mathbf{x}) = \sum_{i=1}^n \left(\sum_{j=1}^i x_j \right)^2$$

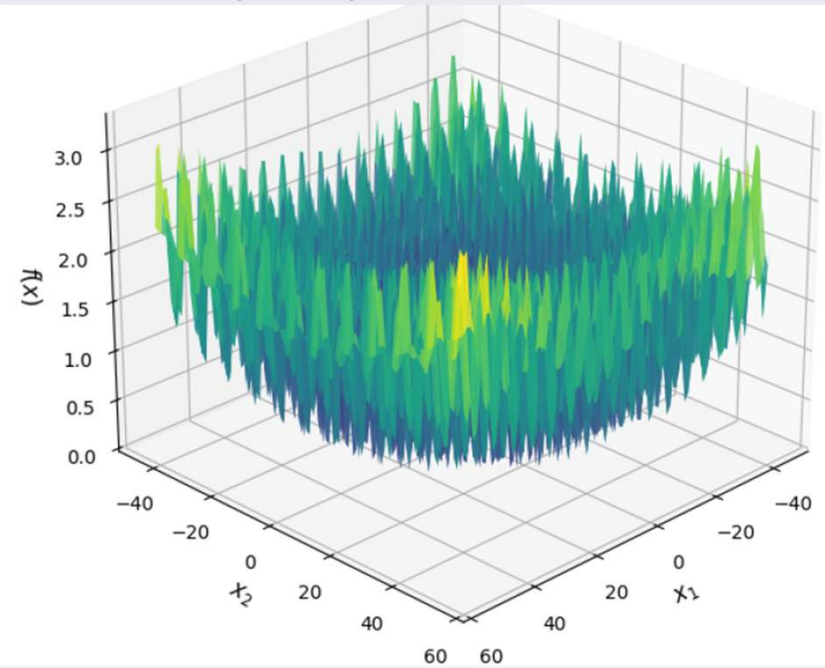
• Domain: $[-4.5, 5.5]^n$



2. Griewank (Multimodal)

$$f(\mathbf{x}) = 1 + \frac{1}{4000} \sum_{i=1}^n x_i^2 - \prod_{i=1}^n \cos\left(\frac{x_i}{\sqrt{i}}\right)$$

• Domain: $[-45, 55]^n$



Benchmark test

Fungsi	Dim	Opt. Global	Mean Best Fitness		Avg. Time (s)		Improvement
			Sobol	Random	Sobol	Random	
Schwefel	3D	0.0	8.30×10^{-5}	1.58×10^{-3}	0.079s	0.055s	+94.75%
	30D	0.0	63.079	118.089	0.323s	0.321s	+46.58%
Griewank	3D	0.0	0.010	0.058	0.050s	0.053s	+82.41%
	30D	0.0	0.821	1.560	0.081s	0.091s	+47.36%

Parameter for the experiment: $N = 20$, $k_{max} = 100$, $r = 0.95$, $\theta = \frac{\pi}{4}$

To ensure a fair comparison, the performance of the Random Initialization was assessed using the average of 100 independent runs to reduce the effect of stochastic variability. In contrast, the Sobol Initialization was evaluated using a single run, as its deterministic nature guarantees consistent results across repeated executions.



Finding Roots of Systems of Nonlinear Equations



Transforming Nonlinear Equations into an Optimization Problem

Consider the system of nonlinear equations

$$g_i(\mathbf{x}) = 0, \quad i = 1, 2, \dots, n,$$

where $\mathbf{x} = (x_1, x_2, \dots, x_n)^T \in D$, and

$$D = [a_1, b_1] \times [a_2, b_2] \times \dots \times [a_n, b_n] \subset \mathbb{R}^n.$$

Define the objective function

$$F(\mathbf{x}) = \frac{1}{1 + \sum_{i=1}^n |g_i(\mathbf{x})|}.$$

so $F(\mathbf{x}) = 1 \iff g_i(\mathbf{x}) = 0, i = 1, \dots, n.$

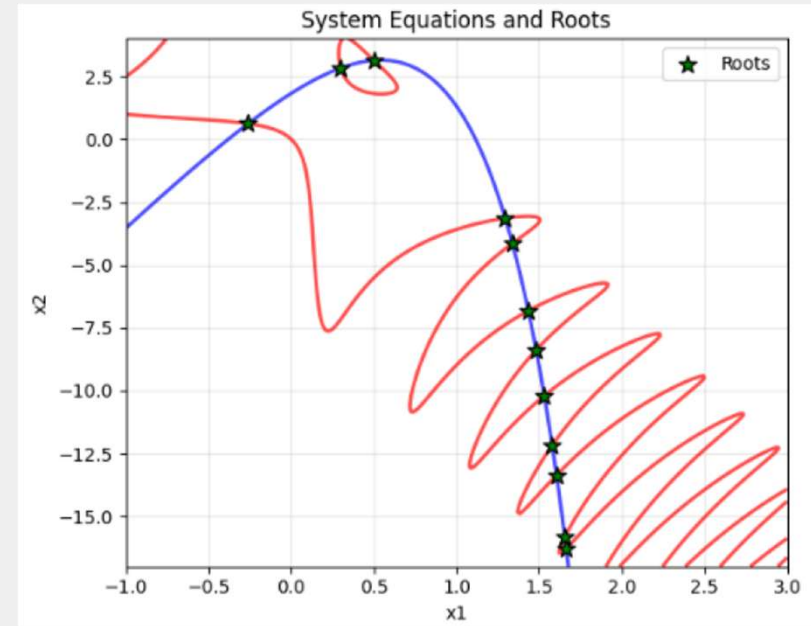
Hence, solving the system of nonlinear equations is equivalent to finding the global maximizer of $F(\mathbf{x})$, whose maximum value is 1.

Benchmark test

$$f_1(x_1, x_2) = 0.5 \sin(x_1 x_2) - 0.25 \frac{x_2}{\pi} - 0.5x_1 = 0$$

$$f_2(x_1, x_2) = \left(1 - \frac{0.25}{\pi}\right) (e^{2x_1} - e) + \frac{ex_2}{\pi} - 2ex_1 = 0$$

dengan $D = \{(x_1, x_2) : -1 \leq x_1 \leq 3, -17 \leq x_2 \leq 4\}$



- The algorithm was executed 100 independent times using different random initializations.
- Each run may converge to a different root.
- Not all roots were found within the 100 runs.
- The computational cost increases proportionally with the number of independent runs.

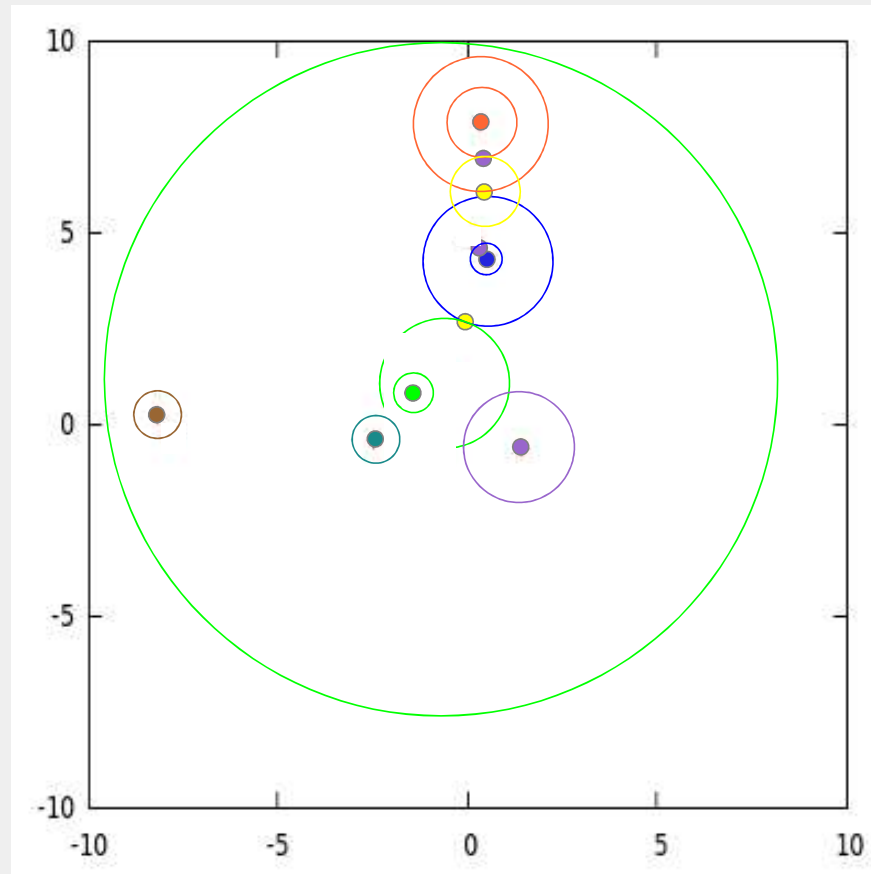
Koordinat Akar (x_i^*)	Frekuensi	Koordinat Akar (x_i^*)	Frekuensi
1. (-0.260599, 0.622531)	11	7. (1.337430, -4.140440)	2
2. (1.530510, -10.202200)	0	8. (0.500000, 3.141590)	36
3. (1.663420, -16.282800)	0	9. (0.299449, 2.836930)	43
4. (1.654580, -15.819200)	0	10. (1.604570, -13.362900)	0
5. (1.433950, -6.820760)	3	11. (1.294360, -3.137220)	2
6. (1.578220, -12.176700)	2	12. (1.481320, -8.383610)	1
Total Eksekusi: 100			



Clustering Method



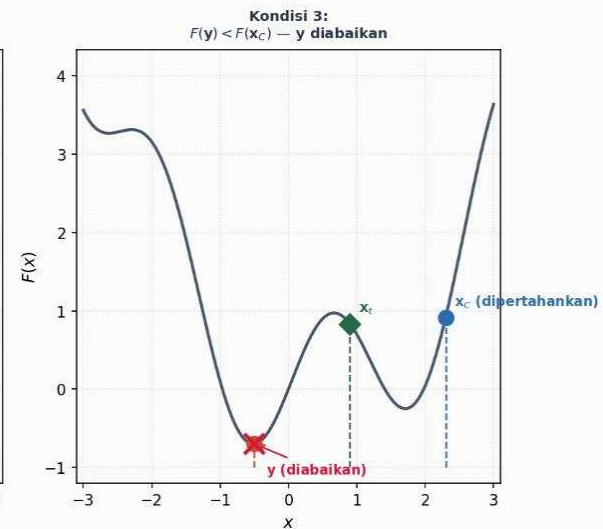
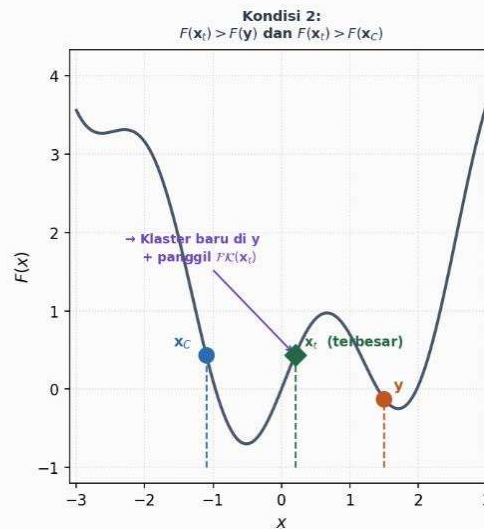
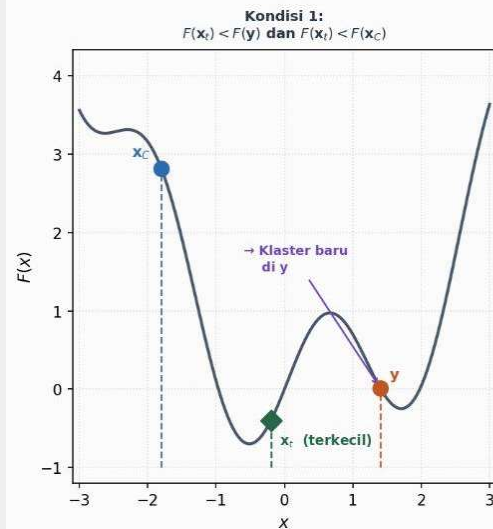
Clustering Idea



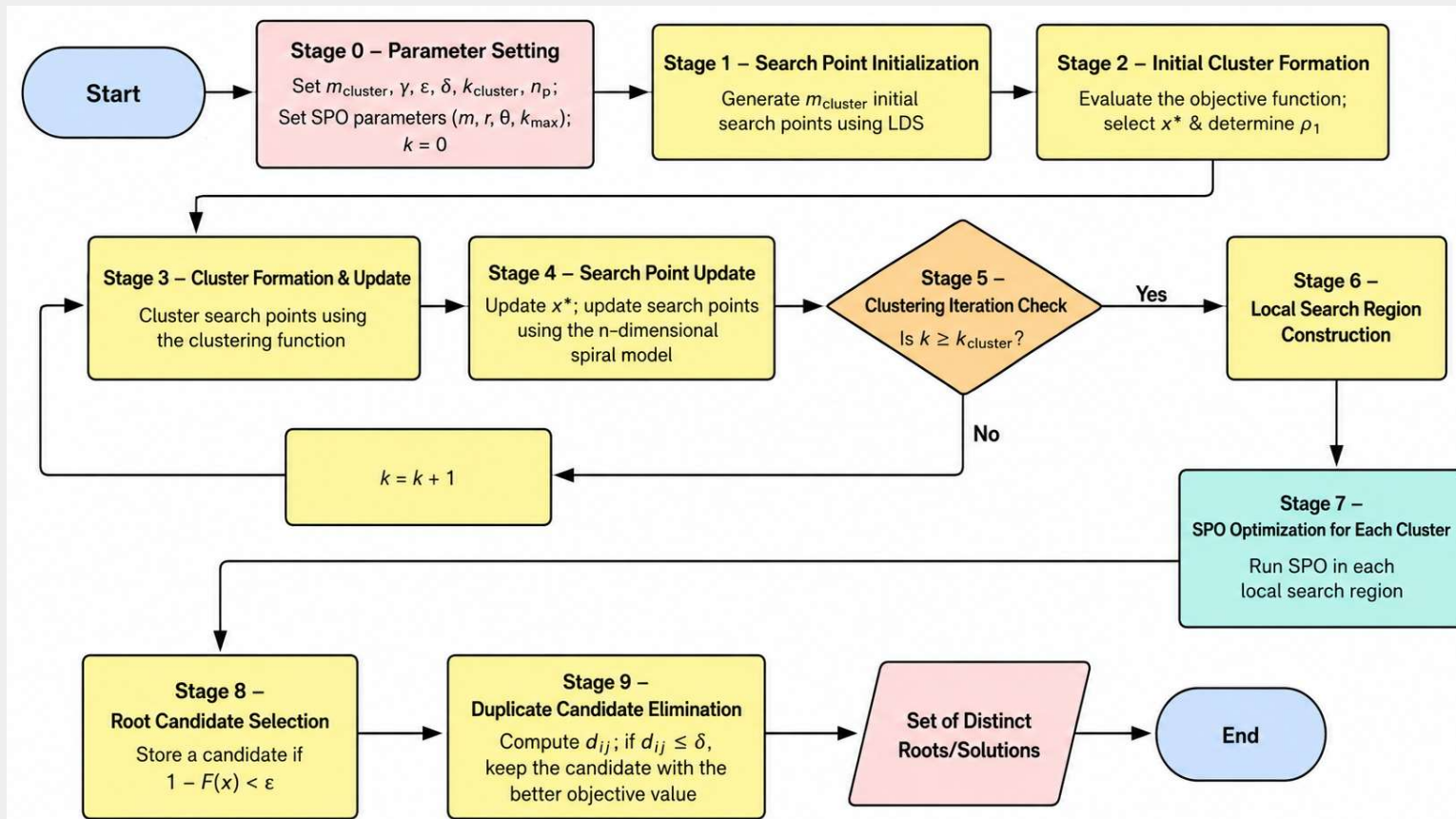
Clustering Idea

Three Decision Conditions

1. If $F(x_t) < F(y)$ and $F(x_t) < F(x_C)$, create a new cluster centered at y with radius $\|y - x_t\|$.
2. If $F(x_t) > F(y)$ and $F(x_t) > F(x_C)$, create a new cluster centered at y , then recursively process x_t .
3. If $F(y) > F(x_C)$, update the cluster center to y .



Clustering Algorithm



Benchmark Test 1

Weierstrass Function

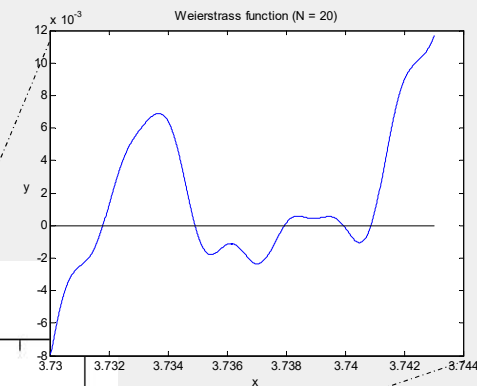
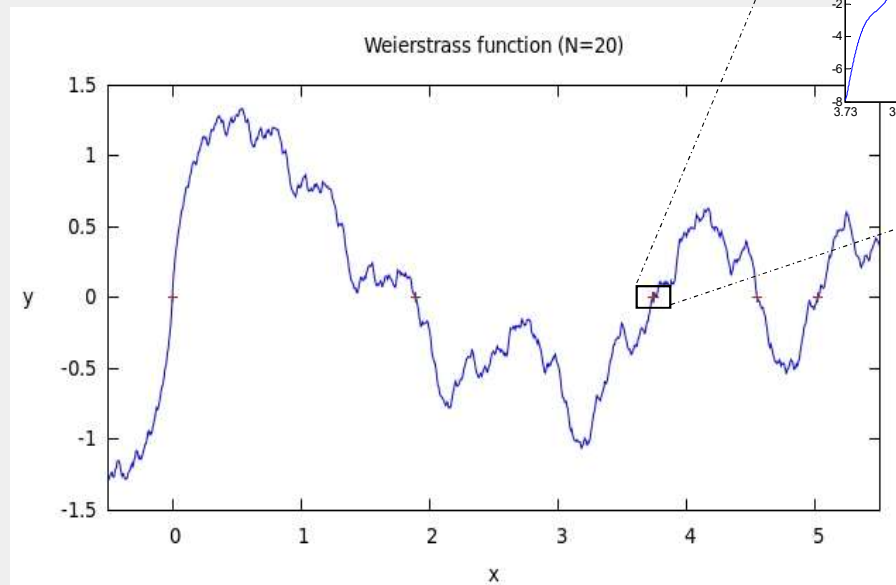
$$f(x) = \sum_{k=1}^{N \rightarrow \infty} \lambda^{(s-2)k} \sin(\lambda^k x)$$

where $1 < s < 2$ and $\lambda > 1$.

Known as a continuous but
nowhere differentiable function

Use $N = 20$, $s = 1.1$, and $\lambda = 1.5$

and domain $-0.5 \leq x \leq 5.5$



Benchmark Test 1

Input

Clustering technique

$$m_{cluster} = 200 \quad \delta = 0.0001$$

$$k_{cluster} = 50 \quad \varepsilon = 0.0000001$$

$$\gamma = 0.9$$

Spiral optimization

$$m = 150 \quad r = 0.95$$

$$k_{max} = 150 \quad \theta = \frac{\pi}{4}$$

Output

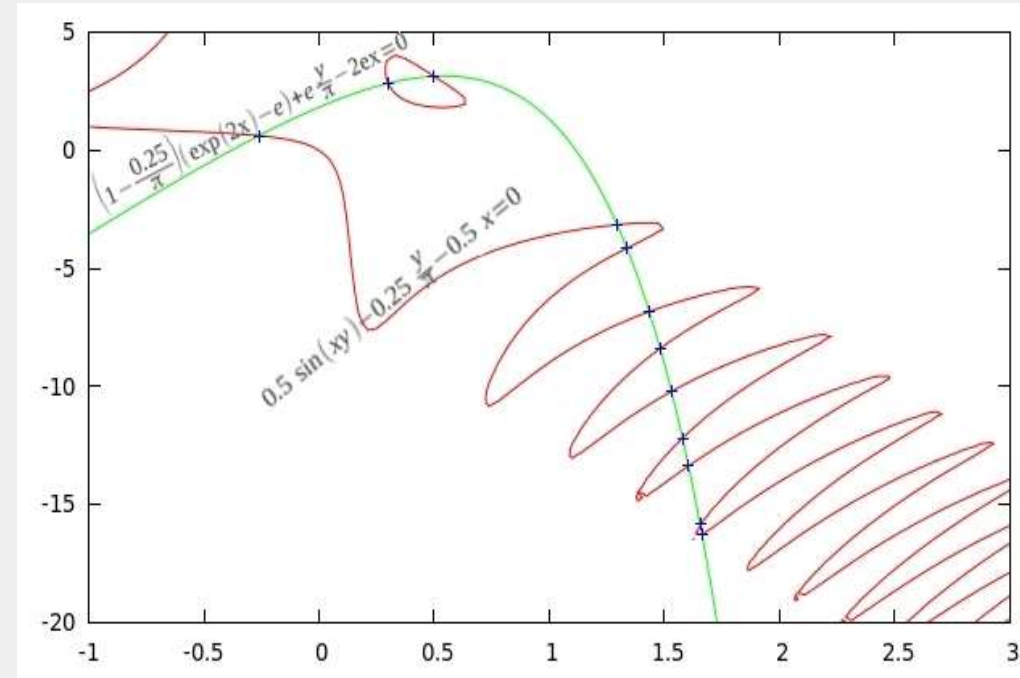
No	x	$g(x)$	No	x	$g(x)$
1	0	0	6	3.73962	9.30683e-08
2	1.88871	-9.77192e-08	7	3.74071	9.10978e-08
3	3.73173	-9.0547e-08	8	4.54986	-9.4056e-08
4	3.73499	-9.69124e-08	9	5.01996	9.56515e-08
5	3.73819	3.67608e-08			

Benchmark Test 2

$$\mathbf{g}(x, y) = \begin{bmatrix} g_1(x, y) \\ g_2(x, y) \end{bmatrix} = \begin{bmatrix} 0.5 \sin(xy) - 0.25 \frac{y}{\pi} - 0.5x \\ \left(1 - \frac{0.25}{\pi}\right)(\exp(2x) - e) + e \frac{y}{\pi} - 2ex \end{bmatrix} = \mathbf{0}$$

search space $-1 \leq x \leq 3, -17 \leq y \leq 4$

Input



Clustering technique

$m_{cluster} = 2000$	$\delta = 0.1$
$k_{cluster} = 10$	$\varepsilon = 0.000001$
$\gamma = 0.3$	

Spiral optimization

$m = 300$	$r = 0.95$
$k_{max} = 300$	$\theta = \frac{\pi}{4}$

Benchmark Test 2

Result

No.	x	y	$F(x, y)$	$g_1(x, y)$	$g_2(x, y)$
1.	-0.260599	0.622531	0.999999	-1.21664e-07	-4.59048e-07
2.	0.299448	2.83693	0.999999	-1.86287e-07	4.69411e-07
3.	0.500001	3.14159	0.999999	-2.47858e-07	-4.77774e-07
4.	1.29436	-3.13722	0.999999	6.26313e-07	2.04202e-07
5.	1.33743	-4.14044	0.999999	7.58506e-07	-8.0146e-09
6.	1.43395	-6.82077	0.999999	-6.48832e-08	-5.50622e-07
7.	1.48132	-8.38361	1	1.96979e-07	1.33501e-07
8.	1.53051	-10.2022	0.999999	-1.98839e-08	5.54021e-07
9.	1.57823	-12.1767	0.999999	3.6169e-07	-4.24333e-07
10.	1.60457	-13.3629	0.999999	2.94067e-07	-2.23847e-07
11.	1.65458	-15.8192	1	-2.27255e-07	2.26023e-07
12.	1.66342	-16.2828	0.999999	-9.24304e-08	-5.61499e-07

Benchmark Test 3

$$\mathbf{g} = \begin{pmatrix} x_1 + \frac{x_2^2 x_4 x_6}{4} + 0.75 \\ x_2 + 0.405 e^{1+x_1 x_2} - 1.405 \\ x_3 - \frac{x_4 x_6}{2} + 1.5 \\ x_4 - 0.605 e^{1-x_3^2} - 0.395 \\ x_5 - \frac{x_2 x_6}{2} + 1.5 \\ x_6 - x_1 x_5 \end{pmatrix} = \mathbf{0}$$

search space $-5 \leq x_i \leq 5$
 $i = 1, 2, \dots, 6$

Input

Clustering technique

$m_{cluster} = 2000$	$\delta = 0.5$
$k_{cluster} = 10$	$\varepsilon = 0.0000001$
$\gamma = 0.001$	

Spiral optimization

$m = 500$	$r = 0.95$
$k_{max} = 500$	$\theta = \frac{\pi}{4}$

Output

1.	$\mathbf{x} = \begin{pmatrix} -1 \\ 1 \\ -1 \\ 1 \\ -1 \\ 1 \end{pmatrix}$	2.	$\mathbf{x} = \begin{pmatrix} -1.04320 \\ -0.550936 \\ 0.431936 \\ 1.75966 \\ -2.10487 \\ 2.19581 \end{pmatrix}$
	$\mathbf{g}(\mathbf{x}) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$		$\mathbf{g}(\mathbf{x}) = \begin{pmatrix} -1.737 10^{-7} \\ -1.50569 10^{-7} \\ -6.54736 10^{-7} \\ 1.49601 10^{-7} \\ -1.61856 10^{-7} \\ 2.95125 10^{-7} \end{pmatrix}$

Complex Roots

Complex number :

$$z = u + vi$$

real part

imaginary part

Complex function : $f(z) = f(u + vi)$

Example :

$$f(z_1, z_2) = \begin{bmatrix} f_1(z_1, z_2) \\ f_2(z_1, z_2) \end{bmatrix} = \begin{bmatrix} z_1^2 + z_2^2 + z_1 + z_2 - 8 \\ z_1 z_2 + z_1 + z_2 - 5 \end{bmatrix} = \mathbf{0} \quad \Rightarrow \quad f(z_1, z_2) = \begin{bmatrix} (u_1^2 - v_1^2 + u_2^2 - v_2^2 + u_1 + u_2 - 8) + (2u_1 v_1 + 2u_2 v_2 + v_1 + v_2)i \\ (u_1 u_2 + u_1 + u_2 - v_1 v_2 - 5) + (u_1 v_2 + u_2 v_1 + v_1 + v_2)i \end{bmatrix} = \mathbf{0}$$

The complex problem can be transformed into real problem :

$$g(x_1, x_2, x_3, x_4) = \begin{bmatrix} g_1(x_1, x_2, x_3, x_4) \\ g_2(x_1, x_2, x_3, x_4) \\ g_3(x_1, x_2, x_3, x_4) \\ g_4(x_1, x_2, x_3, x_4) \end{bmatrix} = \begin{bmatrix} x_1^2 - x_2^2 + x_3^2 - x_4^2 + x_1 + x_3 - 8 \\ 2x_1 x_2 + 2x_3 x_4 + x_2 + x_4 \\ x_1 x_3 + x_1 + x_3 - x_2 x_4 - 5 \\ x_1 x_4 + x_3 x_2 + x_2 + x_4 \end{bmatrix} = \mathbf{0}$$

Benchmark Test 1

Input

Clustering technique

$m_{cluster} = 300$	$\delta = 0.1$
$k_{cluster} = 100$	$\varepsilon = 0.00001$
$\gamma = 0.01$	

Spiral optimization

$m = 100$	$r = 0.95$
$k_{max} = 300$	$\theta = \frac{\pi}{4}$

Output

Root 1	Root 2	Root 3	Root 4
$\mathbf{x} = \begin{pmatrix} 2 \\ 9.19285 \cdot 10^{-7} \\ 1 \\ -2.28927 \cdot 10^{-6} \end{pmatrix}$	$\mathbf{x} = \begin{pmatrix} 1 \\ -1.07511 \cdot 10^{-6} \\ 2 \\ 8.32528 \cdot 10^{-7} \end{pmatrix}$	$\mathbf{x} = \begin{pmatrix} -3 \\ -1.41421 \\ -3 \\ 1.41421 \end{pmatrix}$	$\mathbf{x} = \begin{pmatrix} -3 \\ 1.41421 \\ -3 \\ -1.41421 \end{pmatrix}$
$F(x, y) = 0.999992$	$F(x, y) = 0.999995$	$F(x, y) = 0.999994$	$F(x, y) = 0.999991$
$z = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ real	$z = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ real	$z = \begin{pmatrix} -3 - 1.41421i \\ -3 + 1.41421i \end{pmatrix}$ complex	$z = \begin{pmatrix} -3 + 1.41421i \\ -3 - 1.41421i \end{pmatrix}$ complex

Benchmark Test 2

$$g(z_1, z_2) = \begin{bmatrix} g_1(z_1, z_2) \\ g_2(z_1, z_2) \end{bmatrix} = \begin{bmatrix} z_1^4 + 4z_2^4 - 6 \\ z_1^2 z_2 - 1.6787 \end{bmatrix} = \mathbf{0}$$

with

$$D = \{(z_1, z_2) : -2 \leq u_1, u_2 \leq 2, -2 \leq v_1, v_2 \leq 2\}$$

Input

Clustering technique

$$m_{cluster} = 200 \quad \delta = 0.0001$$

$$k_{cluster} = 50 \quad \varepsilon = 0.0000001$$

$$\gamma = 0.9$$

Spiral optimization

$$m = 150 \quad r = 0.95$$

$$k_{max} = 150 \quad \theta = \frac{\pi}{4}$$

Benchmark Test 2

No	z_1	z_2	$g_1(z_1, z_2)$	$g_2(z_1, z_2)$
1	1.43098-5.08152e-05i	0.819816+6.53159e-05i	-2.6359e-05-1.9781e-05i	4.4030e-05+1.4521e-05i
2	-1.4311+2.82656e-05i	0.819663+3.8692e-05i	3.7359e-05+9.5324e-06i	1.3268e-05+1.2931e-05i
3	1.39602-4.59531e-05i	0.861362+5.32735e-05i	-4.7551e-06+4.4651e-05i	-2.3060e-05-6.6922e-06i
4	-1.39615-8.3548e-05i	0.861226-9.40931e-05i	2.5552e-05-5.2203e-05i	2.5856e-05+1.7507e-05i
5	0.840215+0.840215i	1.00202e-07-1.18897i	-1.0749e-05+7.1284e-06i	2.779e-05-1.7253e-06i
6	0.840205-0.840206i	-1.62053e-06+1.18896i	-1.6171e-05+4.6572e-05i	-1.4268e-05+1.0281e-06i
7	-0.840203+0.84022i	3.68565e-06+1.18896i	-6.6519e-06-2.0486e-05i	1.2003e-05-3.8311e-05i
8	-0.840214-0.840218i	1.17022e-07-1.18897i	-9.5319e-06-1.4893e-05i	3.2477e-05+7.7607e-06i
9	-8.67857e-05-1.43107i	-0.819693-0.000114517i	-6.6203e-05-8.2757e-06i	1.1955e-06+3.0921e-05i
10	0.000159901+1.43101i	-0.819779-0.000211417i	-7.1804e-07-1.073e-05i	3.8877e-05+5.7773e-05i
11	1.8196e-05-1.39627i	-0.861096+1.63072e-05i	-5.6278e-06+3.1534e-05i	5.8816e-05+1.1963e-05i
12	6.66309e-07+1.39611i	-0.861273+2.27387e-06i	4.8289e-05-3.0496e-05i	1.6184e-05-6.0344e-06i



Integer Problem



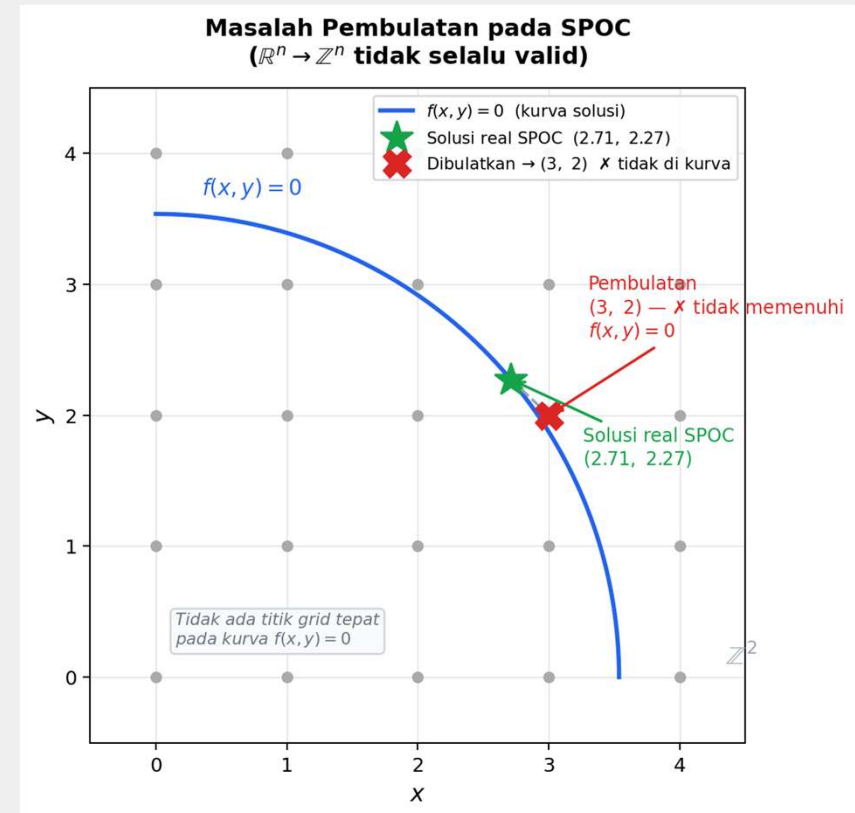
Integer Problems

Challenges

- SPOC produces real-valued solutions. However, many practical problems require integer solutions, such as: Diophantine equations, Scheduling Resource Allocation
- Direct rounding may not produce the correct integer solution.

Modification

- Perform the search in $\mathbb{R}^n$, but evaluate candidate solutions in $\mathbb{Z}^n$ using $q = \text{round}(x)$.
- Extend each variable bound by $\pm 0.5 : \left[l_i - \frac{1}{2}, u_i + \frac{1}{2} \right]$, where l_i is the lower bound and u_i is the upper bound for decision variable x_i . This ensures boundary integers remain reachable after rounding.



Benchmark test 1

SNE

$$15x_1 + 11x_2 - 12 = 0$$

Domain: $x_1, x_2 \in [-50, 50]$

Parameter

m_{cluster}	$= 350$	m_{SPO}	$= 30$
k_{cluster}	$= 10$	k_{maks}	$= 10$
γ	$= 10^{-2}$	r_{SPO}	$= 0,95$
ε	$= 10^{-7}$	θ_{SPO}	$= \pi/4$
δ	$= 0,1$	n_p	$= 3$
r	$= 0,95$		
θ	$= \pi/4$		

Solutions

No.	x_1	x_2
1	-30	42
2	-8	12
3	-19	27
4	25	-33
5	36	-48
6	14	-18
7	3	-3

Benchmark Test 2

SNE

$$x_1^2 + 2^{x_3} \cdot 11^{x_4} = x_2^4$$

Domain: $x_1, x_2, x_3, x_4 \in [1, 20]$

Parameter

m_{cluster}	$= 15000$	m_{SPO}	$= 100$
k_{cluster}	$= 20$	k_{maks}	$= 20$
γ	$= 10^{-2}$	r_{SPO}	$= 0,95$
ε	$= 10^{-7}$	θ_{SPO}	$= \pi/4$
δ	$= 0,001$	n_p	$= 3$
r	$= 0,95$		
θ	$= \pi/3$		

Solutions

No.	x_1	x_2	x_3	x_4
1	0	16	16	0
2	0	1	0	0
3	2	4	0	0
4	7	3	5	0
5	0	4	8	0
6	11	0	4	0

Benchmark Test

Results on Benchmark Problems from [3]:

Case	Ref.	SPOC	Time (s)
1	7	7	0.055
2a	84	1550	33.974
2b	96	4929	116.217
3a	1	1	0.007
3b	1	1	0.005
4 ($n = 3, k = 1$)	7	7	12.474
4 ($n = 3, k = 3$)	11	11	23.591
4 ($n = 4, k = 1$)	5	5	8.215
4 ($n = 4, k = 4$)	5	5	1.494
4 ($n = 5, k = 4$)	5	5	6.184
4 ($n = 6, k = 3$)	6	5	0.226
4 ($n = 7, k = 2$)	5	5	1.088
4 ($n = 8, k = 1$)	3	3	2.028
4 ($n = 9, k = 6$)	4	4	0.887
4 ($n = 10, k = 1$)	3	4	4.102

Case	Ref.	SPOC	Time (s)
5a	7	7	9.853
5b	4	4	45.642
6a	8	13	1.165
6b	5	6	0.813
7	9	9	1.316
8	9	9	0.104
9a	1	1	0.947
9b	1	1	0.221
10	1	1	8.913
11	1	1	0.052
12	1	1	3.014

[3] Sumarti et al. (2023). "SOAC for Diophantine equations". Appl. Soft Comput., 145:110569



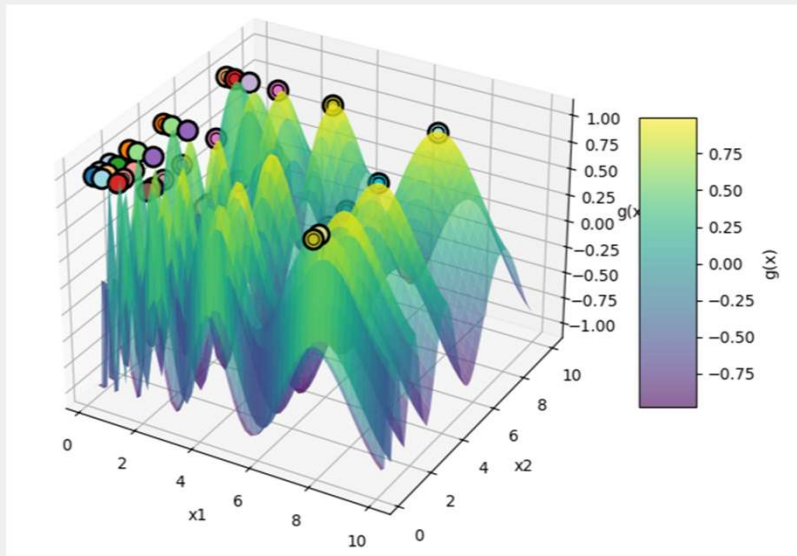
Multimodal Problem



Multimodal Optimization Problem

Multimodal functions may have multiple global (and local) optima, and the objective is to identify all of these solutions.

However, conventional optimization algorithms—including most metaheuristics—typically converge to only one solution per run, depending on the initial starting point.



100× Eksekusi Independen

Maksimum global	36
Berhasil ditemukan	23
Tidak ditemukan	13
Discovery Rate	63.89%
Frekuensi tertinggi	24%

Modification of Clustering Method for Multimodal Problem

Aspect	SNE	Multimodal Optimization
Objective Function	Transform $f(x) = 0$ into $F(x)$.	No transformation; $F(x)$ is evaluated directly.
Target Optimum	Static and absolute: $F(x^*) = 1$.	Dynamic and relative: $f_{\max}/f_{\min}$ depends on the fitness landscape.
Cutoff γ	$F(x_i) > \gamma$.	Relative criterion: $F(x_i) > \gamma F(x_{\text{best}})$.
Local Intensification	SPO converges to points with objective value 1.	SPO converges to peaks (maxima) or valleys (minima).
Termination Criterion	$1 - F(x) < \varepsilon$ (absolute target value).	$F(x \pm c) < F(x)$.
Output	Set of roots ($f(x) \approx 0$).	Set of optimal points (maxima and minima).

The overall algorithmic framework is retained, while the clustering, convergence criterion, and solution selection mechanisms are modified to enable the discovery of multiple optima in multimodal problems.

Benchmark Test 1

Fungsi Objektif $g(x_1, x_2)$

$$g(x_1, x_2) = \frac{1}{2}(x_1^4 - 16x_1^2 + 5x_1) + \frac{1}{2}(x_2^4 - 16x_2^2 + 5x_2)$$

$$\text{Domain } D = [-4, 4]^2$$

Parameter Clustering

- $m_{cl} = 300, k_{cl} = 10$
- $\gamma = -\infty, \delta = 0.1, \epsilon = 10^{-7}$

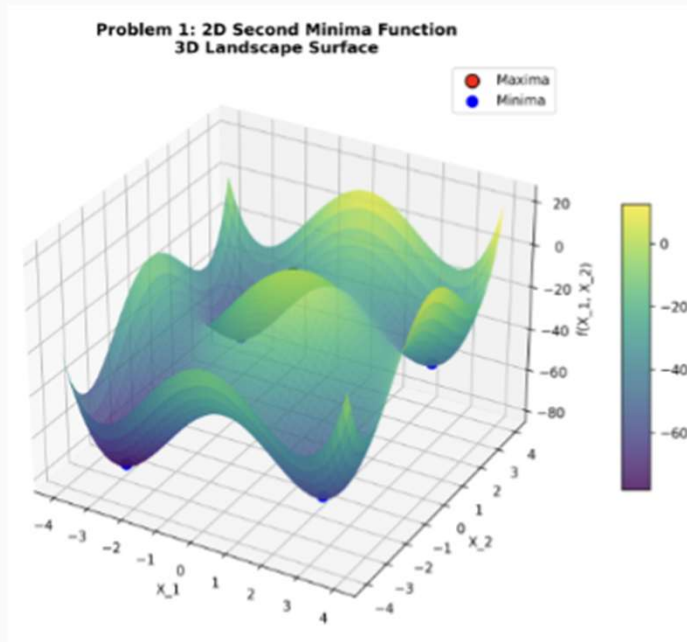
Parameter SPO

- $m = 200, k_{\max} = 230$
- $r = 0.95, \theta = \pi/4$

Benchmark Test 1

Runtime 2.23 s • Cluster 134 (48 Max, 86 Min) • Optimal 5 (1 Max, 4 Min) • Discovery Rate 100%

Runtime: 2.23 s (PySNE) vs 0.65 s (Sidarto et al., 2017), sekitar 3.43× lebih lambat.



Type	x_1	x_2	$g(\mathbf{x})$
Max 1	0.156731	0.156731	0.391225
Min 1	-2.903534	-2.903534	-78.332331
Min 2	-2.903534	2.746803	-64.195612
Min 3	2.746803	-2.903534	-64.195612
Min 4	2.746803	2.746803	-50.058893

Benchmark Test 2

Fungsi Objektif $g(x_1, x_2)$

$$g(x_1, x_2) = \left(4 - 2.1x_1^2 + \frac{x_1^4}{3}\right)x_1^2 + x_1x_2 + (-4 + 4x_2^2)x_2^2$$

$$\text{Domain } D = [-1.9, 1.9] \times [-1.1, 1.1]$$

Parameter Clustering

- $m_{cl} = 300, k_{cl} = 5$
- $\delta = 0.1, \epsilon = 10^{-5}, n_{check} = 2$

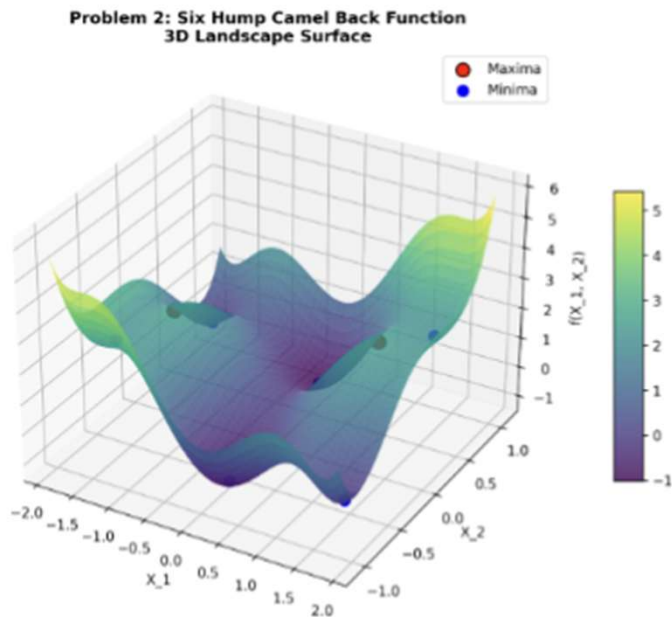
Parameter SPO

- $m = 50, k_{\max} = 250$
- $r = 0.95, \theta = \pi/4$

Benchmark Test 2

Runtime 2.22 s • Cluster 202 (93 Max, 109 Min) • Optimal 8 (2 Max, 6 Min) • Discovery Rate 100%

Runtime: 2.22 s (PySNE) vs 5.14 s (Sidarto et al., 2017), sekitar 2.32× lebih cepat.



Tipe	x_1	x_2	$g(\mathbf{x})$
Max 1	-1.230230	-0.162335	2.496295
Max 2	1.230230	0.162335	2.496295
Min 1	-1.703607	0.796084	-0.215464
Min 2	-1.607105	-0.568651	2.104250
Min 3	-0.089842	0.712656	-1.031628
Min 4	0.089842	-0.712656	-1.031628
Min 5	1.607105	0.568651	2.104250
Min 6	1.703607	-0.796084	-0.215464

Benchmark Test 3

Fungsi Objektif $g(\mathbf{x})$, n -dimensi

$$g(\mathbf{x}) = \sum_{i=1}^n (x_i^2 - 10 \cos(2\pi x_i) + 10)$$

Kasus 2D: Domain $D = [-1, 1]^2$

Parameter Clustering

- $m_{cl} = 500$, $k_{cl} = 10$
- $\delta = 0.1$, $\epsilon = 10^{-6}$, $n_{check} = 1$

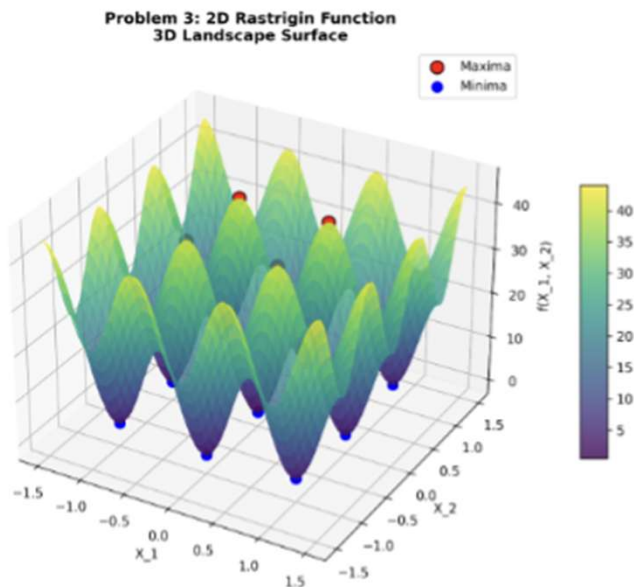
Parameter SPO

- $m = 200$, $k_{\max} = 300$
- $r = 0.95$, $\theta = \pi/4$

Benchmark Test 3

Runtime 3.40 s • Cluster 126 (26 Max, 100 Min) • Optimal 13 (4 Max, 9 Min) • Discovery Rate 100%

Runtime maksimum: 3.40 s (PySNE) vs 0.53 s (Sidarto et al., 2017), sekitar $6.42\times$ lebih lambat.



Tipe	x_1	x_2	$g(\mathbf{x})$
Max 1–4	± 0.5025	± 0.5025	40.5025
Min 1	-0.994959	-0.994959	1.989918
Min 2	-0.994959	0.000000	0.994959
Min 3	-0.994959	0.994959	1.989918
Min 4	0.000000	-0.994959	0.994959
Min 5	0.000000	0.000000	0.000000
Min 6	0.000000	0.994959	0.994959
Min 7	0.994959	-0.994959	1.989918
Min 8	0.994959	0.000000	0.994959
Min 9	0.994959	0.994959	1.989918

4 maks global = seluruh kombinasi ± 0.5025 .

Benchmark Test 4

Fungsi Objektif $g(\mathbf{x})$

$$g(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n \sin(10 \ln x_i)$$

Domain positif $D = [0.25, 10]^2$

Parameter Clustering

- $m_{cl} = 1000, k_{cl} = 10$
- $\gamma = 0.2, \delta = 0.01, \epsilon = 10^{-5}, n_{check} = 3$

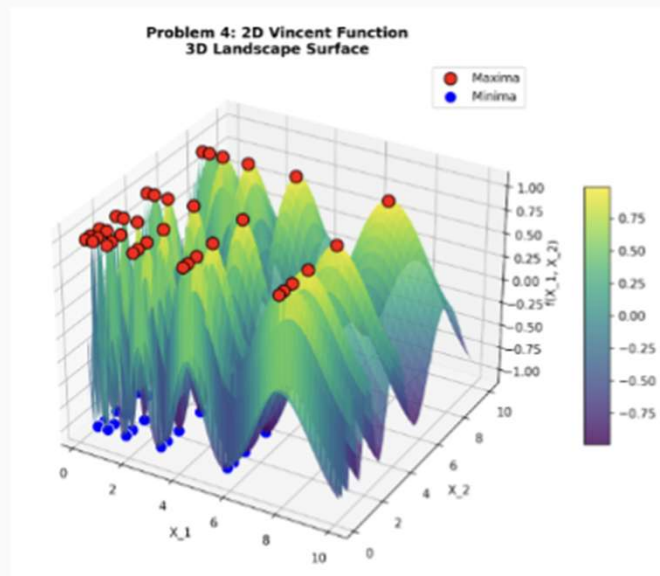
Parameter SPO

- $m = 150, k_{\max} = 150$
- $r = 0.95, \theta = \pi/4$

Benchmark Test 4

Runtime 7.54 s • Cluster 837 (473 Max, 364 Min) • Optimal 61 (36 Max, 25 Min) • Discovery Rate 100%

Runtime maksimum: 3.924 s (PySNE) vs 4.34 s (Sidarto et al., 2017), sekitar 10% lebih cepat.



- 36 maksimum global (6^2) dengan $g(\mathbf{x}) = 1.0$.
- 25 minimum dengan $g(\mathbf{x}) = -1.0$.
- Seluruh puncak rapat di tepi domain ter-isolasi dalam satu kali eksekusi.

Koordinat lengkap 61 titik pada Lampiran.

Benchmark Test 5

Fungsi Objektif $g(\mathbf{x})$

$$g(\mathbf{x}) = \prod_{k=1}^n \left[\sum_{j=1}^5 j \cos((j+1)x_k + j) \right]$$

Kasus 2D: Domain $D = [-10, 10]^2$

Parameter Clustering

- $m_{cl} = 2000, k_{cl} = 15$
- $\gamma = 0.2, \delta = 0.1, \epsilon = 10^{-6}, n_{check} = 1$

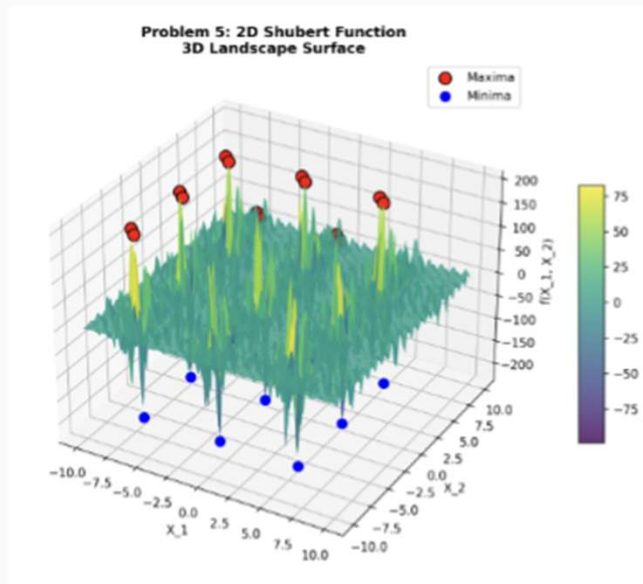
Parameter SPO

- $m = 100, k_{\max} = 500$
- $r = 0.95, \theta = \pi/4$

Benchmark Test 5

Runtime 32.47 s • Cluster 510 (364 Max, 146 Min) • Optimal 27 (18 Max, 9 Min) • Discovery Rate 100%

Runtime maksimum: 22.994 s (PySNE) vs 111.66 s (Sidarto et al., 2017), sekitar 4.86× lebih cepat.



- 18 maksimum global dengan $g(x) = 186.7309$.
- 9 minimum global dengan $g(x) = -210.4823$.
- Solusi tersebar periodik mencakup seluruh domain lebar.

Koordinat lengkap 27 titik pada Lampiran.

THANK YOU



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